Biology IA student handbook

What the Cell · DP Biology · internal assessment. The online version, with tools and tick boxes, is at whatthecell.org/internal-assessment.html

What it is	One scientific investigation of your own, written up as a single report
Marks	4 criteria × 6 marks = 24, worth 20% of your final Biology grade
Time	About 10 hours of investigation, spread over a 16-week route
Length	Up to 3,000 words. Tables, graphs, captions, equations, calculations, headings, citations, references and the AI declaration are not counted
Types	Laboratory experiment, field study, database analysis, or simulation or model
Not needed	Cover page, contents page, abstract. Page 1 shows the title, candidate code and word count
Feedback	Short comments at Tasks 1–6, then one round of full written feedback on your complete draft
Marked by	Your teacher, then checked by IB moderators

Where the marks are. Half the marks (Conclusion and Evaluation) come from the last two sections, which students often rush. Plan your time so that Week 8 is protected for writing them.

The four criteria in plain words

Criterion	What the examiner wants to see	Where you build it
Research design /6	A focused question set in its biology; choices that give enough relevant data, each one explained; a method someone else could repeat	Stages 1–5
Data analysis /6	Clear, precise tables and graphs; uncertainty taken seriously; the right processing and statistics, done accurately	Stages 6–9
Conclusion /6	An answer to your question that your analysis fully supports, explained with biology and compared with published work	Stage 10
Evaluation /6	Specific weaknesses in your design, how much each one matters, and realistic improvements that would fix them	Stage 11

Moving up a band almost always means moving from **stating** (1–2) to **describing** (3–4) to **explaining or justifying** (5–6). The top band always needs a reason, not just a description.

The golden rules

- Start one working document on Day 1 and keep everything in it: notes, sources, photos, raw data and drafts.
- Data collection takes two to three times longer than you expect. Living organisms take longer still, so start plants, cultures and seeds first.

- Never write *proves*. Data *supports* or *suggests*.
- Write *I*, never *we* or *our*: this is an individual report.
- Write scientific names in italics: *Escherichia coli* in full the first time, then *E. coli*.
- Cite every source and declare every use of AI as you go.
- If you are stuck at any stage, ask your teacher straight away.

Your 16-week route

Stage	What you do	Hand in	Week	My due date
1	Choose a research question and write a proposal	Task 1 · Checkpoint 1	1	
2	Mark a mentor IA's research design (in class)	Task 2	2	
3	Write your research design: 25% draft	Task 3 · Checkpoint 2	3	
4	Plan and collect your data	—	3–5	
5	Update the method and add raw data: 50% draft	Task 4 · Checkpoint 3	5	
6–7	Descriptive statistics and graphs	—	6	
8–9	Inferential statistics and the 75% draft	Task 5 · Checkpoint 4	7	
10–11	Write the conclusion and evaluation	Task 6	8	
12	Compile the complete draft and self-assess	Task 7 · Checkpoint 5	9	
13	Revise, format and submit	Task 8 · Checkpoint 6	4 weeks after feedback	

How feedback works. Tasks 1–6 get short comments so you can keep moving. Your teacher gives full written feedback once, on the complete draft (Task 7), within 3 weeks. You then have 4 weeks from the day you receive it to revise. There is no second full review, so make Task 7 your best work.

Academic integrity

- **Your own work.** You may collect data with classmates, but your question, analysis and writing must be your own. Say in your report which data was shared.
- **Cite as you go.** Every source, image, protocol, video and calculator you used goes in your works cited, with the URL and date accessed.
- **Declare AI.** Name the tool, what you used it for, the prompts, and how you used what it gave you. Undeclared AI use is treated as academic misconduct.

- **Stay anonymous.** No names (including yours), school name, photos of faces or exact locations in the report.
- **Ethics first.** No harm to vertebrates; human participants give informed consent (parental consent under 16); microorganisms follow the school's microbiology rules; fieldwork leaves the site as you found it.

Stage 1: Choose your research question

Week 1 · Toddle Task 1 · Checkpoint 1

Your research question shapes every other mark. A narrow, measurable question about a biological phenomenon makes the method, analysis and conclusion straightforward.

Step 1: Pick an investigation type

Type	You will…	Biology examples
Laboratory experiment	Change one variable and measure its effect	Enzyme activity, osmosis, photosynthesis, respiration, microbial growth
Field study	Sample organisms in their environment	Quadrats or transects along a gradient (light, distance from water, trampling)
Database analysis	Extract and analyse published biological data	Gene or protein sequences, species distribution records, health data
Simulation or model	Analyse outputs of a validated model	Population genetics, predator–prey or epidemic models

Step 2: Build the question

Experiment: How does [independent variable, and how it is changed] across [range, number of levels, units] affect [dependent variable], measured by [method/instrument, units], in [organism (Genus species) or biological system]?

Correlation or field study: What is the relationship between [variable 1, units] and [variable 2, units] for [organism (Genus species)] at [type of site] during [season/month]?

Database: What is the relationship between [variable 1] and [variable 2] in [system], using [database] data from [date range]?

Check that someone who has not seen your work can tell, from the question alone:

- ☐ the independent variable, its levels and units, and how it is changed
- ☐ the dependent variable, its units, and how it is measured
- ☐ the organism, with its full scientific name in italics
- ☐ for field studies, the type of location and time of year; for databases, the date range
- ☐ that the question is about a biological phenomenon

Step 3: Compare weak and strong questions

Weak	Strong
Does temperature affect enzymes?	How does temperature (20, 30, 40, 50, 60 °C) affect the rate of catalase activity in *Solanum tuberosum* tissue, measured by the volume of oxygen released from 3% hydrogen peroxide in 60 s?

Weak	Strong
Do plants grow better with more light?	How does light intensity (5 levels, 50–250 μmol m^{-2} s^{-1}) affect the rate of photosynthesis of *Cabomba caroliniana*, measured by change in dissolved oxygen over 10 min?
What is the effect of salt on potatoes?	How does sucrose concentration (0.0–1.0 mol dm^{-3} in 0.2 steps) affect the percentage change in mass of *Solanum tuberosum* cylinders after 24 h?
Does light affect plants in the forest?	What is the relationship between light intensity (lux) and percentage cover of *Pteridium aquilinum* along a woodland transect in the dry season?

Step 4: Check it is feasible

- ☐ At least **5 levels** of the independent variable, across a justified range, or at least **20 data pairs** for a correlation
- ☐ At least **5 trials** per level (needed for standard deviation)
- ☐ Organisms, chemicals and equipment are available; living material can be grown or obtained in time
- ☐ It can be done in roughly 10 hours of investigation time
- ☐ It is safe and ethical: no harm to vertebrates; human participants give informed consent; microorganisms follow school microbiology rules

> **Database tip.** Broad indicators such as national development or wealth indices give weak, messy data. Choose specific biological variables.

What to submit (Task 1)

Draft research question: [...]
Investigation type: [lab / field / database / simulation]
Organism or biological system (Genus species): [...]
Why this interests me (2–3 sentences): [...]
The biology behind it (3–4 sentences, one cited source): [...]
Independent variable and planned levels (with units): [...]
Dependent variable and how I will measure it (with units): [...]
Three variables I will need to control: [...]
Equipment, organisms or data source I need: [...]
Safety, ethical or environmental concerns: [...]
AI declaration: [tool, prompts, how used – or 'No AI used']

Your teacher will **approve, refine or redirect** your question. Do not start practical work until it is approved.

Stage 2: Learn from mentor IAs

Week 2 · Toddle Task 2 (repeated in class at Stages 6, 10 and 11)

The fastest way to learn what moderators reward is to mark real Biology IAs yourself. You will do this four times, once for each criterion, just before you write that section of your own report.

Session	When	You mark	Then you write
Research design	Week 2	Question, context, method	Your 25% draft
Data analysis	Week 6	Tables, graphs, uncertainties, statistics	Your analysis
Conclusion	Week 8	Five sample conclusions, ranked	Your conclusion
Evaluation	Week 8	Five sample evaluations, ranked	Your evaluation

How each session works (30 minutes)

1. **Read and annotate on your own (10 min).** For each rubric aspect, decide a band and note the evidence. Is it **stated**, **described** or **explained**? What is missing?
2. **Agree as a group (10 min).** Agree a group mark or ranking. Write 3–5 sentences giving the band for each aspect, quoting evidence, naming one strength and one flaw that decided it.
3. **Score reveal (10 min).** Your teacher shares the moderator's mark and comments. Where did you differ, and why?

What to look for

- **Research design:** a specific question with units, method and *Genus species*; a context narrowed to the variables; every methodological choice justified; a method you could repeat.
- **Data analysis:** titles, headers, units, consistent decimal places; uncertainties in headers; error bars defined; statistics chosen for a reason; worked calculations.
- **Conclusion:** answers the question with real values; avoids *proves*; uses SD, error-bar overlap and p-values; explains the biology and compares with published work.
- **Evaluation:** weaknesses tied to specific apparatus, biological assumptions or steps, not "human error"; explains which way each flaw biased the data; realistic, detailed improvements.

What to submit (Task 2)

Aspect	Group band	Evidence (quote or page)	One change that would raise the mark
Research question in context			
Methodological considerations			
Description of method			

Moderator’s mark vs ours: […]
What the moderator focused on: […]
Three ideas I will use in my own IA: 1. … 2. … 3. …
One mistake I will avoid: […]

Stage 3: Research design — the 25% draft

Week 3 · Toddle Task 3 · Checkpoint 2

This draft builds **Research design**. It is your *planned* method; you will update it later to match what you actually did. The top band needs every decision **explained**, so keep asking: *why this choice?*

3.1 Research question

State it at the top of the report, using the Stage 1 checklist.

3.2 Scientific context (description of the system)

- Name the biological topic or phenomenon (for example, enzyme denaturation, osmosis, the light-dependent reactions).
- Describe the relevant background biology, **narrowed to your two variables**, not the whole topic.
- Explain how your variables relate to the biological system.
- Add a figure (diagram or photo) if it helps explain a mechanism.
- Cite academic papers or textbooks for every biological claim.

3.3 Theory of direct relevance and hypothesis

- Explain the known effect of your independent variable on your dependent variable and the **biological mechanism** behind it.
- Write a prediction of the trend you expect, justified by that mechanism. Do **not** write it as null/alternative hypotheses; those come in Stage 8.
- Describe the expected pattern (for example, "rate increases to an optimum near 40 °C, then falls as the enzyme denatures").
- Cite sources for the hypothesis.

3.4 Variables

Variable	Variable and units	How it is changed, measured or controlled	Why (biological effect if uncontrolled)
Independent			
Dependent			
Control 1			
Control 2			
Control 3			
Monitored (cannot be controlled)			

Independent variable

- ☐ Identified with units; at least 5 levels and the range stated
- ☐ Levels and range justified by a pilot trial or research
- ☐ Positive or negative control groups explained, if used (for example, boiled enzyme as a negative control)
- ☐ Method of setting it explained and justified over alternatives
- ☐ Database or simulation: how it was selected, why it is suitable, the link, and why the data is reliable
- ☐ A captioned photo of the independent variable (screenshots for databases)

Dependent variable

- ☐ Quantifiable, with units, and its choice justified (any derived value, such as rate or percentage change, identified)
- ☐ Measuring method explained and justified over alternatives (for example, gas syringe vs counting bubbles)
- ☐ Number of trials stated and justified (at least 5 per level; 20+ pairs for a correlation)
- ☐ Validity measures explained: calibrated instruments, clean glassware, one stock solution for all trials, a standard mixing method, one person doing each task, a standard viewing angle, the same equipment throughout
- ☐ A captioned photo of the dependent variable

Controlled variables

- ☐ At least three, each with the biological effect it would have if uncontrolled (for example, temperature changes enzyme kinetic energy)
- ☐ How each was kept constant **and monitored**
- ☐ Any confounding variables that cannot be controlled but are monitored
- ☐ A captioned photo showing control

3.5 Apparatus, materials and uncertainties

- List every item with brand, size and quantity. Give concentrations (mol dm^{-3}, g dm^{-3} or %) and volumes (cm^3) or masses (g) for every chemical and biological reagent; dimensions for containers. Metric units only.
- Add a labelled sketch or photo of the set-up.

Instrument	Quantity measured	Uncertainty (±, units)	How the uncertainty was decided

Deciding uncertainty: analogue scales, usually ± half the smallest division; digital readouts, ± the last digit; or the manufacturer's value. Hand timing: allow for reaction time.

3.6 Justification of methods

- Describe any pilot trial and how it improved the method.
- Justify the choice of organism or tissue: why is it suitable?
- Explain instrument choices where options exist (pH probe vs pH paper; colorimeter vs colour chart).
- Explain why random sampling is needed and how you will do it.
- Cite any standard protocol you followed.

3.7 Procedure

- Numbered steps in logical order, detailed enough for someone else to repeat exactly.
- Include how you change the independent variable, measure the dependent variable, maintain and measure controls, sample randomly, take repeats and record qualitative observations.
- Refer to annotated diagrams or photos in the relevant steps.
- Database or simulation: the input selections and the criteria used to extract data.

3.8 Safety, ethics and environment

Hazard	Risk	Control measure	Disposal	Source (SDS or guidance)

Address every point that applies:

- ☐ Chemical, biological and apparatus hazards, with SDS cited for every chemical
- ☐ Eye protection, gloves, lab coat
- ☐ Minimal use of chemicals and organisms
- ☐ Safe disposal of plants, tissues, cultures and chemicals
- ☐ **Human participants:** confidential informed consent; parental consent under 16
- ☐ **Animals:** follow the IB animal experimentation guidelines; no harm to vertebrates
- ☐ **Microorganisms:** source of the culture, aseptic technique, incubation at no more than school-safe temperatures, sterilisation and disposal
- ☐ **Fieldwork:** minimal impact on the site and personal safety
- ☐ If none apply, a sentence saying so

3.9 Works cited and AI declaration

Start your reference list now, including protocols, videos and websites. Your AI declaration states the tool, what you used it for, the prompts, and how you used the output.

Self-check before submitting Task 3

- [ ] Context is focused on my two variables and fully cited
- [ ] Hypothesis is justified by a biological mechanism
- [ ] Every variable has a method and a biological reason
- [ ] Range, instruments, organism and number of trials are each justified
- [ ] Every instrument has an uncertainty
- [ ] Procedure is numbered and repeatable, with figures
- [ ] Safety, ethics and environment are complete

Your teacher will **sign off your method** or request specific changes before you collect data.

Stage 4: Plan and collect your data

Weeks 3–5

Biological material is unpredictable: seeds fail, cultures contaminate, enzymes lose activity. Plan for problems, start early, and record everything at the moment you measure it.

4.1 Make your data collection plan (30 minutes)

Write 5–7 sentences answering:

- When will you get your materials and organisms?
- When and where will you set up and run the investigation?
- How many sessions will you need, and how long is each?
- What is your back-up if something fails? (If you are growing plants or cultures, start now.)
- One question you still have about your data collection.

The four requirements of data collection

1. **Enough data:** at least 5 levels and at least 5 trials per level (or 20+ data pairs).
2. **Quantitative and qualitative data.**
3. **Uncertainty and consistent digits:** every value to the precision of its instrument.
4. **Data for every variable:** independent, dependent and controlled, plus photos.

4.2 Part A: Photos

Photograph (without showing your face): the materials; you carrying out the investigation; the independent variable; the dependent variable; the control of variables. Caption each photo to explain what it shows and how it relates to your design. Database or simulation studies: use screenshots.

4.3 Part B: Measurement uncertainty

Tool	Unit	Uncertainty (±)	How the uncertainty was determined
Tool for the independent variable			
Tool for the dependent variable			
Tool(s) for controlled variables			

Be realistic. A stopwatch reads to ±0.01 s, but reaction time is about ±0.2 s. A tape measure on uneven ground is less precise than its markings. Counts (colonies, stomata, organisms in a quadrat) may need an uncertainty based on how much two counters disagree. Databases: report the stated uncertainty, or explain that an unknown uncertainty is itself a limitation.

4.4 Part C: Quantitative raw data

Raw data is what you read from the instrument, before any calculation.

[IV] / [unit] (±)	Trial 1 [DV] / [unit] (±)	Trial 2	Trial 3	Trial 4	Trial 5	Observations

- ☐ Numbered, specific title ("Table 2: Mass of *Solanum tuberosum* cylinders before and after 24 h in six sucrose concentrations")
- ☐ Independent variable in the first column
- ☐ Units and uncertainty in every header
- ☐ Same number of decimal places for every value of a variable
- ☐ All levels and trials
- ☐ A separate table of control variable readings showing they stayed within an acceptable range

Date and time	Control variable / unit (±)	Value	Within acceptable range?

4.5 Part D: Qualitative data

Record observations for each level: colour of tissue or indicator, bubbling, turgidity, contamination, leaf condition, weather and site conditions. Use annotated photos, sketches or maps. Database studies: state that no qualitative data was available.

4.6 Method change log

Date	What changed	Why	Effect on data

Collaboration: you may collect data with others, but your write-up and analysis must be your own; state which data was shared. **Anomalies:** never delete a strange value. Flag it, note what happened, and deal with it in your analysis.

Stage 5: The 50% draft

Week 5 · Toddle Task 4 · Checkpoint 3 · About 1,500 words

Combine your research design with your raw data and update the method so it describes what you **actually did**. Your teacher will confirm whether your data is sufficient or whether you need to collect more while there is time.

Section	Status
Research question, context, hypothesis	From Stage 3, refined
Variables, apparatus, uncertainties, justification of methods	From Stage 3, **updated**
Procedure	**Updated** to match what you did (use your change log)
Safety, ethics and environment	From Stage 3
Raw data: quantitative, qualitative, control readings	**New**
Works cited and AI declaration	**Updated**

Formatting from now on

- ☐ Title on page 1 reflecting the research question; **no cover or contents page**
- ☐ Word count and candidate code on page 1 (leave the code blank if you do not know it yet)
- ☐ Anonymous: no names (including yours), schools or specific locations
- ☐ *I*, not *we* or *our*
- ☐ Scientific names in italics with correct capitals
- ☐ Every table and figure numbered, captioned, referred to in the text, placed next to where it is mentioned, sharp and inside the margins
- ☐ Tables do not break across pages (or repeat the header)
- ☐ Consistent in-text citations; reference list with URLs and dates accessed

Raw data checklist

- ☐ Tables have specific titles and are organised by group or data type
- ☐ Independent variable in the first column
- ☐ Data for every level and trial in the method (for very large data sets, a representative sample)
- ☐ Control variable data included
- ☐ Qualitative data organised by level of the independent variable

Self-check before submitting Task 4

- [ ] Method and procedure updated to what I actually did
- [ ] Raw data tables complete and formatted
- [ ] Photos captioned and referred to
- [ ] About 1,500 words; works cited and AI declaration updated

Stage 6: Descriptive statistics

Week 6 · Begins with the in-class mentor session on data analysis.

Descriptive statistics summarise your data so a reader can see what it shows; they do not test your hypothesis (that is Stage 8). Choose the statistics that fit **your** question and data, not what a classmate did.

For every calculation, write in your report:

1. **Why** you did it.
2. **Where** to find the results ("See Table 4 for the mean and standard deviation at each sucrose concentration.").
3. **One worked example** using your data: the formula with your numbers, or a clear, cited screenshot of a spreadsheet or online calculator, labelled with the level it uses.

Round calculated values to the same number of decimal places as the least precise raw value.

Step 1: Preliminary calculations (if needed)

Do these **for each trial**, before averaging.

Calculation	Formula	Biology example
Total change	final − initial	Change in height of seedlings
Percentage change	(final − initial) ÷ initial × 100	Change in mass of potato cylinders in osmosis
Rate	change ÷ time, or 1/t	Volume of O_2 per minute; 1/time for a colour change
Other	e.g. Simpson's diversity index, percentage cover	Field studies

Step 2: What kind of study is it?

- **Levels of an independent variable with repeated trials** → Step 3.
- **Paired data, no repeats** (e.g. light intensity and leaf area for 30 plants) → Step 6.
- **Unique data** (presence/absence in quadrats, DNA or protein sequences) → Step 7.

Step 3: Check the shape of the data (skew)

Calculate skew **for each level** using the SKEW function in Excel or Google Sheets, or a cited online calculator.

Absolute skew	Meaning	Use
0 to 0.5	Approximately symmetrical	Mean
0.5 to 1.0	Slightly skewed	Mean
1.0 to 2.0	Moderately skewed, acceptable	Mean

Absolute skew	Meaning	Use
Above 2.0	Strongly skewed	Median

Negative skew = more values above the mean; positive skew = more values below. Report direction and strength for each level. Check outliers with the 1.5 × IQR rule; justify keeping or removing them and, if it matters, show results both ways.

Step 4: Central tendency

Your data	Use	Spreadsheet
Quantitative, most skews within ±2.0	**Mean** (most students)	=AVERAGE( )
Quantitative, most skews beyond ±2.0	**Median**	=MEDIAN( )
Qualitative, ranked	**Median**	=MEDIAN( )
Qualitative, categories	**Mode**	=MODE( )

Step 5: Spread

You used	Trials per level	Use	Spreadsheet
Mean	5 or more	**Standard deviation (SD)**	=STDEV( )
Mean	30 or more	SD or **standard error (SE)**	=STDEV( )/SQRT(n)
Mean	4 or fewer	**Range**	=MAX( ) − MIN( )
Median	5 or more	**Interquartile range (IQR)**	=QUARTILE(,3) − QUARTILE(,1)
Median	4 or fewer	**Range**	=MAX( ) − MIN( )

About 68% of values lie within 1 SD of the mean. A small SD suggests consistent, precise data; a large SD suggests high natural variation or error. Biological material often varies a lot, so comment on it.

Summary table

Level of IV / unit	Skew	Mean (or median) / unit	SD (or IQR/range) / unit	n
Level A				
Level B				
Level C				
Level D				
Level E				

Step 6: Correlation studies

1. Draw a scatter graph.
2. **Linear** → Pearson's r. **Curved but always rising or always falling (monotonic)** → Spearman's rank r_s.
3. Calculate with all data pairs (=PEARSON( ) or a cited calculator).
4. Describe direction (positive or negative) and strength: up to 0.5 weak; 0.5–0.8 moderate; 0.8–1.0 strong.

Step 7: Unique data

Write a paragraph and build a summary table describing the data (for example, the number of quadrats in which each species was present, or sequence similarity percentages).

Stage 7: Present your data in graphs

Week 6

Graph your **processed** data (means, medians, frequencies), not every raw value. The graph follows from what you calculated in Stage 6. Excel handles custom SD error bars more easily than Google Sheets; clearly scanned hand-drawn graphs are also acceptable.

Choose your graph

You calculated	Independent variable is	Graph	Biology example
Mean + SD (or range)	Categories	**Bar chart** with ± 1 SD error bars	Rate of respiration in three yeast species
Mean + SD (or range)	Numbers	**Points of the means** with ± 1 SD error bars and a trendline	Catalase activity at five temperatures
Median + IQR	Either	**Box and whisker** plot	Leaf length in sun vs shade (skewed data)
Mode (qualitative)	Levels of a variable	**Stacked bar chart** of frequencies, with a key	Seed colour phenotypes in different conditions
Mode (qualitative)	No levels	**Pie chart**, with a key	Proportions of species in a sample
Correlation (r)	Paired data	**Scatter graph** with a trendline	Light intensity vs percentage cover along a transect
Unique data	—	Suitable visual (e.g. phylogenetic tree) with a descriptive paragraph	Sequence comparison between species

Every graph must have

- ☐ A numbered, descriptive title: "Figure 5: Mean percentage change in mass of *S. tuberosum* cylinders in six sucrose concentrations (error bars show ± 1 SD)"
- ☐ Independent variable on the x-axis, dependent variable on the y-axis
- ☐ Axis labels with metric units and proportional scales
- ☐ Error bars, defined in the caption
- ☐ A key, where colours are used
- ☐ A size that makes it easy to read

Choose a trendline (numerical data)

Trendline	Shape	Biology example
Linear	Steady increase or decrease	Osmosis across a concentration range

Trendline	Shape	Biology example
Polynomial	Rises then falls (a peak)	Enzyme activity vs temperature or pH
Exponential	Rapid, continual increase	Early bacterial or yeast population growth
Logarithmic	Rises quickly then levels off	Photosynthesis vs light intensity (saturation)

Report **R^2**: the closer to 1, the better the line fits. Use it to justify your trendline, or explain why you did not add one.

Write about each graph (3–5 sentences)

1. Where: "See Figure [n] for …"
2. Type: "A [graph type] was used because [justification linked to the data]."
3. Comparison: "Mean [DV] increased from [value] at [level] to [value] at [level] …"
4. Error bars: "The SD at [level] was large relative to the mean, suggesting high biological variability …"
5. Overlap: "The error bars for [level] and [level] overlap, suggesting the difference may not be real …"

(Trendline) "A [type] trendline (R^2 = [value]) shows [shape]; the slope is steep until [x], then levels off …"

Stage 8: Inferential statistics

Week 7

Inferential statistics tell you whether your results are likely to be **real** or could have happened **by chance**. Choose the test that fits your data; running a test that does not fit loses marks.

Step 1: Key ideas

- **Significant:** very unlikely to be due to chance. **Not significant:** could be due to chance.
- **p-value:** the probability that results like yours happen by chance alone (p = 0.025 means a 2.5% chance).
- **Significance level:** the cut-off used in biology is **0.05**. If $p < 0.05$, the result is significant.
- Significance is more likely with a large difference between groups, small spread within groups, and a large sample.

In your report, define an inferential test, *significant* and *not significant*, and state your significance level, with a citation.

Step 2: Choose your test from your graph

Your graph and data	Test	Statistic
Bar chart of means ± SD, **2** groups, unrelated samples	Unpaired t-test (two-tailed)	t
Bar chart of means ± SD, **2** groups, same samples before and after	Paired t-test	t
Means ± SD, **3 or more** groups	One-way ANOVA + Tukey post-hoc	F
Box and whisker of medians (skewed data)	Kruskal–Wallis + Dunn post-hoc	H
Stacked bar or pie chart of counts; phenotype ratios; species association	Chi-squared	χ^2
Scatter or line of means, **linear** trendline	Pearson's correlation	r
Scatter or line of means, **curved but monotonic**	Spearman's rank correlation	r_s
Line of means, **non-monotonic** (e.g. enzyme optimum)	ANOVA + post-hoc	F

Explain **why** your test suits your data. Chi-squared works only on counts, not percentages or means.

Step 3: Statistical hypotheses

These are different from your research hypothesis. Match the wording to the test.

Test	Null hypothesis (H_0)	Alternative hypothesis (H_1)
t-test, ANOVA, Kruskal–Wallis	There is no significant **difference** in [DV] between [levels of IV]; any difference may be due to chance.	There is a significant **difference** in [DV] between [levels of IV].
Chi-squared (association)	There is no significant **association** between [species A] and [species B] at [site].	There is a significant association between [species A] and [species B].
Chi-squared (goodness of fit)	There is no significant difference between observed and expected [counts/ratio].	There is a significant difference between observed and expected [counts/ratio].
Pearson, Spearman	There is no significant **correlation** between [X] and [Y].	There is a significant correlation between [X] and [Y].

Step 4: Run the test and show working

Use a spreadsheet or a reputable online calculator. Include a clear, captioned, cited screenshot or worked calculation. If ANOVA or Kruskal–Wallis is significant, report the post-hoc test to show **which** groups differ.

Step 5: Results table

Test	n	Test statistic	p-value	Significant at 0.05?

Round to 2–3 decimal places; number the table; one value per cell.

Step 6: Accept or reject the null hypothesis

I used the [test] to determine whether the results were statistically significant.
The calculated [statistic] was [value] and the p-value was [value].
Because the p-value was [lower/higher] than the 0.05 significance level,
the results [are/are not] significant, so I [reject/fail to reject] the null hypothesis.

Stage 9: The 75% draft

Week 7 · Toddle Task 5 · Checkpoint 4 · About 2,000 words

Add your analysis from Stages 6–8 to the 50% draft. Your teacher will return a short list of priority gaps to fix before the complete draft.

Descriptive analysis

- [ ] Preliminary calculations, if needed, with a worked example
- [ ] Skew or normality check, with its justification
- [ ] Central tendency and spread, each justified
- [ ] Correlation coefficient, if relevant
- [ ] Numbered, titled table of results by level of the independent variable
- [ ] Worked examples or cited screenshots, large enough to check, labelled with the level used

Data presentation

- [ ] Titled, numbered graphs with labelled axes, units and proportional scales
- [ ] Graph type justified
- [ ] Means, medians or frequencies compared across levels
- [ ] Error bars described relative to the central value and to each other (spread, reliability, overlap)
- [ ] Trendline justified with R^2; trend described (slopes, optimum, plateau, inflection points)

Inferential analysis

- [ ] Test chosen and justified
- [ ] Null and alternative hypotheses stated
- [ ] Cited calculation or screenshot
- [ ] Results table: statistic, n, p-value compared with 0.05
- [ ] Sentence accepting or rejecting the null hypothesis

Throughout

- [ ] Uncertainties in every raw data header
- [ ] Consistent decimal places
- [ ] Works cited and AI declaration updated

Self-check before submitting Task 5

- [ ] Every number in the analysis can be traced back to my raw data
- [ ] Every calculation, graph and test has a stated reason
- [ ] About 2,000 words

Stage 10: Write the conclusion

Week 8 · Toddle Task 6 (part A) · Begins with the in-class session ranking five sample conclusions.

The conclusion answers your research question from your analysis and explains it with biology.

Level	What it looks like	Max
State	A brief answer with no detail	2/6
Describe	A detailed account of the pattern, but no reasons why	4/6
Justify	Data **and** biological reasons for why it happened	6/6

Component 1: The conclusion statement

- Answer the research question directly, describing the relationship from your data.
- State how far your research hypothesis is supported: fully, partly or not at all.
- Be cautious: *the data suggests*, *indicates*, *supports*. Never *proves*. Do not generalise beyond your organism, range or conditions.
- Correlation studies: discuss correlation versus causation.

Component 2: Support from your analysis

- Quote specific values: means, differences between levels (marginal or pronounced), comparison with a control.
- Describe features of the data: optimum, plateau or saturation point, threshold, lag or exponential phase, inflection point, isotonic point (osmosis), positive or negative and monotonic or not.
- Use spread: narrow SD suggests high precision; wide SD suggests natural variation or error; compare error-bar sizes and overlap.
- Use the inferential test: did you reject the null hypothesis, and what does that say about chance?

Component 3: Uncertainty and error analysis

- **Systematic errors** shift all values one way and affect **accuracy** (for example, a balance reading 0.02 g high; blotting potato cylinders more on one side).
- **Random errors** scatter values and affect **precision** (for example, natural variation between leaves; reaction time when timing).
- Do not list careless mistakes (“I spilled some solution”) as errors.
- Judge how measurement uncertainty, variation (outliers, skew, SD, IQR, R^2) and the p-value affect your **confidence** in the conclusion.

Component 4: Biological context

- Restate the biological phenomenon you investigated.

- Explain the biological processes that produced your results (for example, collision frequency and active-site denaturation; water potential gradients; photon capture by photosystem II).
- Explain the variability using biology (for example, genetic variation between individual plants).
- If results were unexpected, explain the process you expected and suggest why it did not occur.
- Compare with published studies or accepted values, with citations; where an accepted value exists, calculate percentage error: |experimental − accepted| ÷ accepted × 100.

Worked example

The effect of light intensity on the rate of photosynthesis in Elodea.

> As light intensity increased from 500 to 2,000 lux, the mean volume of oxygen produced rose from 4.2 cm^3 to 18.6 cm^3, a strong positive correlation (Pearson's r = 0.82). The linear trendline in Figure 7 fits the data well (R^2 = 0.67). The Pearson's test gave p = 0.02, below 0.05, so the relationship is statistically significant and unlikely to be due to chance. This supports the hypothesis that oxygen production increases with light intensity. More photons are available at higher intensity, activating more electrons in photosystem II and accelerating the photolysis of water, which releases oxygen (Figure 10).

Every claim has a number, a statistic or a biological reason.

Conclusion frame

1. Answer: The data suggests that as [IV] increased from [x] to [y], [DV] [increased/decreased/levelled off] from [value] to [value].
2. Hypothesis: This [supports / partly supports / does not support] my hypothesis that [...].
3. Pattern: [size of change, optimum, plateau, threshold, overlap of error bars].
4. Significance: The [test] gave [statistic] and p = [value], so the result [is/is not] significant.
5. Uncertainty: [Systematic/random errors, SD, R^2] mean my confidence is [high/moderate/low] because [...].
6. Biology: This is explained by [mechanism] (citation).
7. Comparison: My result [agrees/differs] with [published study or value] (percentage error [z]% if relevant), because [...].

Self-check

- [ ] First sentence answers the research question with data
- [ ] Hypothesis support stated; *suggests* or *supports*, never *proves*
- [ ] Descriptive and inferential results used as evidence
- [ ] Errors and uncertainty discussed in terms of confidence
- [ ] Results and variability explained with biology and compared with cited published work

Stage 11: Write the evaluation

Week 8 · Toddle Task 6 (part B) · Begins with the in-class session ranking five sample evaluations.

Errors vs weaknesses. Errors (in your conclusion) are deviations from the true value during measurement. Weaknesses (in your evaluation) are flaws in the **design** of the investigation. They are not careless mistakes or hypothetical events with no evidence.

Level	What it looks like	Max
State	Generic weaknesses and improvements	2/6
Describe	Specific weaknesses with relevant improvements	4/6
Explain	The **relative impact** of each specific weakness, with improvements explained	6/6

Component 1: Identify methodological weaknesses (aim for at least three)

- **Sample size:** 5–14 trials is very small, 15–30 small, over 30 large. Small samples hide trends and magnify anomalies, especially with variable biological material.
- **Levels and range:** could you see the full biological trend (for example, the optimum or the isotonic point)? What would extra levels have shown?
- **Control of variables:** which factors fluctuated (temperature, light, age or source of organisms, tissue thickness)?
- **Apparatus precision:** was it sensitive enough? (Measuring 1 cm^3 with a 100 cm^3 cylinder is a limitation.)
- **Duration:** was there enough time for the biological response (growth, osmotic equilibrium, culture growth)?
- **Data source:** for databases and simulations, how reliable is it?

Support each weakness with **evidence from your data**: an unexpected trend, inconsistent replicates, a large SD at one level, an outlier.

Component 2: Evaluate the impact

- Explain **how** it affected your ability to draw a conclusion.
- Judge it **major** (could change the conclusion) or **minor** (added some variability).
- Say whether it pushed results **higher or lower**, or increased scatter.
- Rank your weaknesses by impact.

Component 3: Realistic improvements

- One improvement per weakness, directly targeting it.
- Specific: not “better equipment” but “a digital pH probe (± 0.01) instead of universal indicator paper” or “a cork borer and scalpel guide to cut cylinders to 2.0 cm”.

- Feasible in a school laboratory or field site.
- Explain **how** it would reduce error, increase precision or improve validity.

Write confidently: avoid “I may have…” or “it could have…”.

Worked example

Weakness	Impact	Improvement
The water bath was set to 30 °C, but the hot plate cycled on and off, so the temperature fluctuated between 25 °C and 35 °C during each 10-minute trial.	**Major.** Enzyme activity in yeast respiration is highly temperature-sensitive, so the fluctuations caused inconsistent respiration rates between trials. This explains the large SD at 30 °C and makes it hard to tell whether the trend was caused by the intended temperature.	Replace the hot plate with a thermostatically controlled digital water bath (± 0.5 °C). The stable temperature would reduce variation between trials, giving more precise data and a smaller SD.

Evaluation table to complete

Weakness (design, with evidence from my data)	Impact (major/minor, direction, why)	Specific, realistic improvement and how it helps

Self-check before submitting Task 6

- [ ] At least three design weaknesses, none of them careless mistakes
- [ ] Each backed by evidence from my data
- [ ] Each judged major or minor, with direction and reasons, and ranked
- [ ] Each has a specific, school-feasible improvement explaining how it helps
- [ ] Conclusion and evaluation both added to my draft

Stage 12: Compile the complete (100%) draft

Week 9 · Toddle Task 7 · Checkpoint 5 · Maximum 3,000 words

This is the draft that receives full written feedback, returned within 3 weeks. It must be complete and your best work. You cannot submit it until Tasks 1–6 are done.

Report structure

Title (reflects the research question)
Candidate code · Word count

1. Research question
2. Scientific context, theory and hypothesis
3. Variables
4. Methodology

- Apparatus, materials and uncertainties
- Justification of methods
- Procedure (what you actually did)
- Safety, ethical and environmental considerations

5. Results

- Raw data (or a representative sample) and control variable data
- Qualitative observations

6. Data analysis

- Descriptive analysis with worked examples
- Graphs and their description
- Inferential analysis

7. Conclusion
8. Evaluation
9. Works cited
10. AI declaration

Before you submit

- **Check the thread:** question, method, data, conclusion and evaluation refer to the same variables, organism, range and units.
- **Cut to the limit:** remove general background and repetition. Tables, graphs, captions, equations, calculations, headings, in-text citations, references and the AI declaration do not count.
- **Write in paragraphs**, not bullet points or question-and-answer form.

Self-assessment

Mark your draft with the score sheet in the Appendix. Your teacher compares your marks with theirs; the gaps show where to focus.

Criterion	My mark /6	Evidence (section, page)	What would lift it one band
Research design			
Data analysis			
Conclusion			
Evaluation			
Total /24			

Self-check before submitting Task 7

- [ ] Every section present, in paragraphs
- [ ] Method matches what I did
- [ ] Under 3,000 words, count on page 1
- [ ] All formatting checks from Stage 5 pass
- [ ] References complete and cited in the text; AI declaration complete
- [ ] Self-assessment completed with evidence

Stage 13: Revise, format and submit

Toddle Task 8 · Checkpoint 6 · Due 4 weeks after you receive feedback

13.1 Revise using your feedback

1. Read all feedback first and sort it by criterion.
2. Fix the comments that affect the most marks first.
3. A comment about missing justification or uncertainty may need changes in several sections.
4. Record every change in a revision log.

Teacher comment	Criterion	What I changed	Section and page

13.2 Final formatting

Word count: maximum 3,000 words. Not counted: charts and diagrams, captions, equations and calculations, numbers in tables, in-text citations, the reference list, the AI declaration and headers. Over the limit? Cut repetition.

Layout

- ☐ No title page; title on page 1 with candidate code and word count
- ☐ One font throughout (for example Arial), size 11 or 12, never below 10
- ☐ Spacing between 1 and 2; margins no smaller than about 1.3 cm
- ☐ Tables numbered (“Table 1: …”); photos, diagrams and graphs numbered with captions (“Figure 1: …”)
- ☐ No table split across pages unless the title and header repeat
- ☐ Typically 6–12 pages

Writing

- ☐ Past tense for what you did; *I*, never *we*
- ☐ Spelling checked; scientific names in italics (*Escherichia coli*, then *E. coli*)
- ☐ Paragraphs, not outlines or question-and-answer
- ☐ Every table and figure referred to in the text
- ☐ Anonymous: no names, photos of you, school or specific locations

Citations

- ☐ One consistent style (MLA or APA, as your teacher specifies)
- ☐ Websites with URL and date accessed

- [ ] Images, graphs and screenshots you did not make are cited
- [ ] AI declaration: tool, purpose, prompts, how output was used

13.3 Submit

- [ ] Every teacher comment appears in my revision log
- [ ] Student declaration on the score sheet signed
- [ ] Uploaded to Toddle as a **PDF**

Appendix A: Biology IA score sheet

Field	Entry
Candidate code	
Session	
Investigation title	
Word count	
Investigation type	Laboratory experiment / Field study / Database analysis / Simulation or model

Student declaration

- [ ] This is my own work and this is the final version.
- [ ] I have acknowledged the words and ideas of others, whether written, oral or visual.
- [ ] My use of AI followed academic integrity guidelines, and I have declared all of it.

Signed: _____ Date: _____

Teacher declaration

- [ ] I can authenticate that the work was performed by the student as described.

Signed: _____ Date: _____

Criterion	Aspect 1	Aspect 2	Aspect 3	Criterion mark
Research design	RQ context __/6	Methodological considerations __/6	Description of method __/6	__/6
Data analysis	Communication __/6	Uncertainty __/6	Processing __/6	__/6
Conclusion	Conclusion __/6	Scientific context __/6		__/6
Evaluation	Weaknesses / limitations __/6	Improvements __/6		__/6
Raw total				**__/24**

Marks are awarded on a best-fit basis across the aspects of each criterion.

What each band looks like, in plain words

These are our own summaries to help you self-assess. Your teacher marks with the official IB descriptors.

Research design

Marks	Question and context	Planning for good data	Method
1–2	The question is there, but the biology around it is missing	You name some things that matter for the data	Someone else could not repeat it
3–4	The question sits in a general area of biology	You describe how you will get relevant data and enough of it	Someone could repeat it, with a few gaps or guesses
5–6	The question sits in a focused, fitting biological context	You explain why each choice gives relevant and sufficient data	Someone could repeat it exactly

Data analysis

Marks	Communication	Uncertainty	Processing
1–2	Hard to follow and imprecise	Uncertainty barely considered	Some useful processing, but big gaps or mistakes
3–4	Clear or precise, not both	Uncertainty considered, with real gaps or mistakes	Useful processing, with real gaps or mistakes
5–6	Clear and precise	Uncertainty handled properly	Useful processing, done properly and accurately

A *big* gap stops you reaching a valid conclusion. A *real* gap still lets you conclude, but limits how valid or detailed the conclusion is.

Conclusion

Marks	Your conclusion	Link to published science
1–2	Relevant, but your analysis does not back it up	A surface-level comparison
3–4	Described, but does not fully match your analysis	Some useful comparison
5–6	Justified and fully matches your analysis	Justified by a relevant comparison

Evaluation

Marks	Weaknesses	Improvements
1–2	General weaknesses are named	Realistic improvements are named
3–4	Specific weaknesses are described	Realistic, relevant improvements are described
5–6	You explain how much each specific weakness matters	You explain how each realistic improvement helps

Appendix B: Key terms

Term	Meaning
State / Outline / Describe	Brief answer / brief summary / detailed account (lower and middle bands)
Explain / Justify	Detailed account with reasons or causes / valid reasons or evidence (top band)
Independent (manipulated) variable	The variable you deliberately change
Dependent (responding) variable	The variable you measure
Controlled variable	Kept constant so it does not affect the result
Confounding variable	Cannot be controlled; monitor and report it
Raw data	Values read directly from the instrument
Uncertainty	The range within which the true value lies (±)
Precision / Accuracy	How close repeats are to each other / how close a value is to the true value
Random / Systematic error	Unpredictable scatter (reduces precision) / consistent shift (reduces accuracy)
Validity / Reliability	Measures what it claims to / gives the same result when repeated
Standard deviation (SD)	Spread of values around the mean
Interquartile range (IQR)	Spread of the middle half of the data
Skew	How lopsided a distribution is compared with a normal curve
Correlation coefficient (r)	Strength and direction of a relationship, −1 to +1
R^2	How well a trendline fits the data, 0 to 1
Null hypothesis (H_0)	No significant difference, association or correlation
p-value	The probability that results like yours occurred by chance
Significant	$p < 0.05$
Percentage error	|experimental − accepted| ÷ accepted × 100

Always check the details with your teacher: they follow the current IB Biology guide.

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